

Maharashtra HSC Exam, 2017

Chemistry

Time-3 Hours

M.M- 80

General Instructions:

1. All questions are compulsory.
2. Answers to the questions of Section-I and Section –II should be written in the same answer book.
3. Draw neat, labelled diagrams and write balanced chemical equations wherever necessary.
4. Figures to the right indicate full marks.
5. Use of logarithmic table is allowed.
6. Every new question must be started on a new page.

Section-I

Question 1: Select and write the most appropriate answer from the given alternatives for each sub- questions:

[7]

(i) An antifriction alloy made up of antimony with tin and copper, which is extensively used in machine bearings is called:

- (a) Duralumin (b) Babbitt metal (c) Spiegeleisen (d) Amalgams

(ii) Which of the following pairs is an intensive property?

- (a) Density, viscosity (b) Surface tension, mass (c) Viscosity, internal energy
(d) Heat capacity, volume

(iii) Fe^{2+} ions react with nitric oxide formed from reduction of nitrate and yields a brown coloured complex:

- (a) $[\text{Fe}(\text{CO})_5\text{NO}]^{2+}$ (b) $[\text{Fe}(\text{NH}_3)_5\text{NO}]^{2+}$ (c) $[\text{Fe}(\text{CH}_3\text{NH}_2)_5\text{NO}]^{2+}$
(d) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$

(iv) MnO_2 and $\text{Ca}_3(\text{PO}_4)_2$ present in iron ore get reduced to Mn and P in the zone of:

- (a) combustion (b) reduction (c) fusion (d) slag formation

(v) An ionic compound crystallises in FCC type structure with 'A' ions at the centre of each face and 'B' ions occupying corners of the cubes. The formula of compound is:

- (a) AB_4 (b) A_3B (c) AB (d) AB_3

(vi) On passing 1.5F charge, the number of moles of aluminium deposited at cathode are:

- (a) 1.0 (b) 0.5 (c) 4 (d) 1

Solution 1:

- (i) (b) Babbitt metal
- (ii) (a) Density, viscosity
- (iii) (d) $[\text{Fe}(\text{H}_2\text{O})_5\text{NO}]^{2+}$
- (iv) (b) Reduction
- (v) (b) A_3B
- (vi) (c) 0.50
- (vii) (b) 0.5

Question 2: Answer any six of the following:

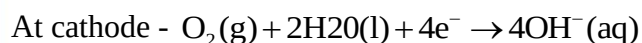
[12]

- (i) What are 'fuel cells'? Write cathode and anode reactions in a fuel cell.
- (ii) Derive the relationship between half-life and rate constant for first order reaction.
- (iii) Explain magnetic separation process of ores with the help of a neat, labelled diagram.
- (iv) Derive the relationship between relative lowering of vapour pressure and molar mass of solute.
- (v) Define the term 'enthalpy'. What will happen to the internal energy if work is done by the system?
- (vi) Nitrogen does not form pentahalides. Give reason.
- (vii) Calculate the percentage efficiency of packing in case of simple cubic cell.
- (viii) Write the electronic configuration of the following elements:
 - (a) Sulphur ($Z=16$)
 - (b) Krypton ($Z=36$)

Solution 2:

- (i) Fuel cells are those cells which produce electrical energy from the combustion of fuels such as hydrogen, carbon monoxide or methane e.g. $\text{H}_2 - \text{O}_2$ cell.

Cell reaction:



- (ii) For a first order reaction,

$$t = \frac{2.303}{k} \log \frac{[R]_0}{[R]}$$

Where, $[R]_0$ is initial concentration and $[R]$ is the final concentration.

At

$$t = t_{1/2} [R] = \frac{[R]_0}{2}$$

$$t_{1/2} = \frac{2.303}{k} \log \frac{[R]_0}{[R]_0 / 2}$$

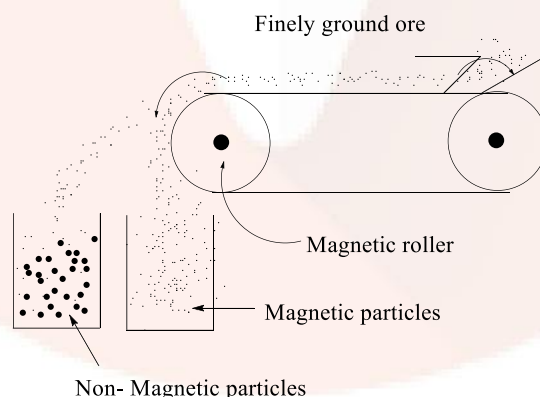
$$= \frac{2.303}{k} \log 2$$

$$t_{1/2} = \frac{2.303}{k} \times 0.3010$$

$$t_{1/2} = \frac{0.693}{k}$$

Therefore, $t_{1/2}$ for a first order reaction is independent of initial concentration.

(iii) Magnetic separation is based on differences in magnetic properties of ore and gangue. The powered ore is dropped on the belt moving on the two rollers, one of which is a strong magnet. The magnetic particles attracted by the magnet falls nearer to the roller whereas non-magnetic particles falls away from the roller. It is used for the separation of magnetic ores like magnetite (Fe_3O_4) and pyrolusite (MnO_2).



(iv) Let W_2 g of solute of molar mass M_2 be dissolved in W_1 g of solvent of molar mass M_1 . Hence, number of moles of solvent, n_1 and number of moles of solute n_2 , in solution are given as,

$$n_1 = \frac{W_1}{M_1} \text{ and } n_2 = \frac{W_2}{M_2}$$

The mole fraction of solute, x_2 is given by,

$$x_2 = \frac{n_2}{n_1 + n_2} = \frac{(W_2/M_2)}{[(W_1/M_1) + (W_2/M_2)]} \dots\dots\dots(i)$$

$$\frac{\Delta P}{P_1^0} = \frac{P_0 - P_1}{P_1^0} = \frac{P_1^0 x_2}{P_1^0} = x_2 \dots\dots\dots(ii)$$

Combining (i) and (ii) equations:

$$\frac{\Delta P}{P_1^0} = \frac{P_0 - P_1}{P_1^0} = x_2 = \frac{(W_2/M_2)}{[(W_1/M_1) + (W_2/M_2)]} \dots\dots\dots(iii)$$

For a dilute solution, $n_1 \gg n_2$.

Thus, n_2 may be neglected in comparison with n_1 in equation (i) and equation (ii), we have,

$$\frac{\Delta P}{P_1^0} = \frac{n_2}{n_1} = \frac{(W_2/M_2)}{W_1/M_1} = \frac{W_2 \times M_1}{W_1 \times M_2}$$

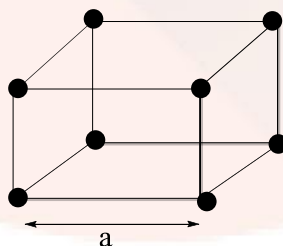
(v) The enthalpy of a system is defined as the sum of the internal energy of the system and the energy that arises due to its pressure and volume.

According to the first law of thermodynamics, internal energy depends on the heat exchanged by the systems and work done by the system or on the system. Taking that no heat is exchanged, the internal energy will decrease. Work done by the system is taken negative and thus $-W$.

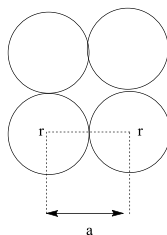
Therefore, internal energy $\Delta U = q - W$.

(vi) Nitrogen has a s and p-orbitals. It does not have d-orbitals to expand its covalency beyond four. Due to which it does not form pentahalides.

(vii) In a simple cubic cell, the edge length of the unit cell = a and radius of the sphere = r .



As spheres touch each other along the edge, therefore $a = 2r$



$$\text{Number of sphere per unit cell} = \frac{1}{8} \times 8 = 1$$

$$\text{Volume of the sphere} = \frac{4}{3} \pi r^3$$

$$\text{Volume of cube} = a^3 = (2r)^3 = 8r^3$$

$$\text{Therefore, packing fraction} = \left(\frac{4}{3} \pi r^3 \right) / 8r^3 = 0.524$$

$$\text{Percentage efficiency} = 52.4\%$$

(viii) (a) Sulphur (Z=16)

$$\text{Electronic configuration} = 1s^2 2s^2 2p^6 3s^2 3p^4$$

(b) Krypton (Z=36)

$$\text{Electronic configuration} = 1s^2 2s^2 2p^6 3s^2 3p^4 3d^{10} 4s^2 4p^6$$

Question 3: Answer any Three of the following:

[9]

(i) How is phosphine prepared using the following reagents?

(a) HCl (b) H₂SO₄ (c) Caustic Soda

(ii) 0.05 M NaOH solution offered a resistance of 31.6 Ω in a conductivity cell at 298K. If the cell constant of the cell is 0.367 cm⁻¹, calculate the molar conductivity of NaOH solution.

(iii) Calculate ΔH° for the reaction between ethene and water to form ethyl alcohol from the following data:

$$\Delta H^\circ \text{C}_2\text{H}_5\text{OH}_{(l)} = -1368 \text{ kJ}$$

$$\Delta H^\circ \text{C}_2\text{H}_5\text{OH}_{(g)} = -1410 \text{ kJ}$$

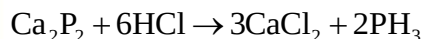
Does the calculated ΔH° represent the enthalpy of formation of liquid ethonal?

(iv) In the Arrhenius equation for a first order reaction, the values of 'A' and 'E_a' are

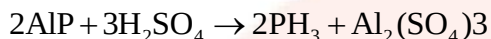
$4 \times 10^{13} \text{ sec}^{-1}$ and 98.6 kJmol^{-1} [$R=8.314 \text{ kJmol}^{-1}$]

Solution 3:

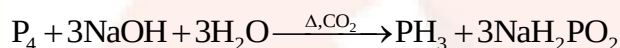
(i) (a) Phosphine can be prepared by the reaction of calcium phosphate with dilute HCl.



(b) Phosphine can be prepared by the decomposition of aluminum phosphide with dilute sulphuric acid.



(c) Phosphine can be prepared by heating white phosphorous with concentrated sodium hydroxide solution in an inert atmosphere by CO_2 .



(ii) Conductivity,

$$k = \frac{\text{Cell constant}}{\text{Resistance}} = \frac{0.367 \text{ cm}^{-1}}{31.6 \text{ ohm}} \\ = 0.0116 \text{ Scm}^{-1}$$

Molar conductivity,

$$\Lambda_m = \frac{k}{C_m} \\ = \frac{0.0116 \text{ Scm}^{-1}}{0.05 \text{ molL}^{-1}} \\ = \frac{0.0116 \text{ Scm}^{-1}}{0.05 \text{ mol} \times (1000 \text{ cm}^3)^{-1}} \\ = 232 \text{ Scm}^2 \text{ mol}^{-1}$$

(iii) $\text{C}_2\text{H}_4 + \text{H}_2\text{O} \rightleftharpoons \text{CH}_3\text{CH}_2\text{OH}$

$$\Delta H^0 = \Delta_c H^0 (\text{products}) - \Delta_c H^0 (\text{reactants}) \\ = -1368 \text{ kJ} - (-1410 \text{ kJ}) \\ = -1368 \text{ kJ} + 1410 \text{ kJ} \\ = 42 \text{ kJ}$$

No, the enthalpy of formation of liquid is not same as the calculated ΔH^0 for the reaction. Therefore, it does not represent the enthalpy of formation of liquid ethanol.

(iv)

$$t_{1/2} = \frac{0.693}{k}$$

$$\Rightarrow k = \frac{0.693}{10 \times 60}$$

$$= \frac{0.693}{600}$$

$$= 0.001155$$

As,

$$\log k = \log A - \frac{E_a}{2.303RT}$$

$$\Rightarrow \log 0.001155 = \log 4 \times 10^{13} - \frac{98.6 \times 10^3}{2.303 \times 8.314T}$$

$$\Rightarrow -2.937 = 13.602 - \frac{98600}{19.147T}$$

$$\Rightarrow T = \frac{98600}{19.147 \times 16.539}$$

$$= \frac{98600}{316.672} = 311.36K$$

Question 4: State Faraday's first law of electrolysis:

Write any 'two' uses of each of the following:

- (a) H₂SO₄, (b) Chlorine

Distinguish between crystalline solids and amorphous solids.

A solution of a substance having 1.8×10^{-3} kg has the osmotic pressure of 0.52 atm at 280K. Calculate the molar mass of the substance used. [Volume = 1 dm³, R = 8.314 JkK⁻¹ mol⁻¹]

OR

Define the following:

- (a) Leaching (b) Metallurgy (c) Anisotropy

Derive an expression for maximum work.

The boiling point of benzene is 353.23K. When 1.80 gram of non-volatile solute was dissolved in 90 grams of benzene, the boiling point is raised to 354.11 K. Calculate the molar mass of solute. [K_b for benzene = 2.53 K mol^{-1}]

Solution 4:

Faraday's first law of electrolysis states that "the amount of any substance deposited or liberated at the electrode is directly proportional to the quantity of electricity passing through the electrode.

(a) Uses of H_2SO_4 :

- (i) In the manufacture of fertilizers like ammonium sulphate and super phosphate, etc.
- (ii) In petroleum refining, manufacture of pigments, paints, dye stuff, intermediates.
- (iii) In detergent industry.

(b) Uses of Chlorine:

- (i) Bleaching of cotton and textiles.
- (ii) Extraction of gold and platinum.
- (iii) Sterilizing drinking water.

Difference between crystalline solids and amorphous solids:

S.No.	Property	Crystalline Solids	Amorphous Solids
1.	Shape	Definite shape and characteristics geometry.	Irregular shape and lack characteristic geometry.
2.	Melting Point	Sharp melting Point	Do not have sharp melting point and melt gradually over a range of temperature.
3.	Nature	True solids	Pseudo solids
4.	Heat of Fusion	Definite	Indefinite

Osmotic pressure,

$$\pi = 0.52 \text{ atm}, T = 280 \text{ K}, W_B = 1.8 \times 10^{-3} \text{ kg} = 1.8 \text{ g}, V = 1 \text{ dm}^3 = 1 \text{ L}, R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\begin{aligned}
 M_B &= \frac{W_B RT}{\pi V} \\
 &= \frac{1.8 \times 8.314 \times 280}{0.52 \times 1} \\
 &= \frac{4190.256}{0.52} \\
 &= 8058.184\text{g} \\
 &= 8.058\text{kg}
 \end{aligned}$$

OR

(a) Leaching is the process of treating the powdered ore with a suitable reagent which can selectively dissolve the ore but not the impurities and further reprecipitated.

(b) Metallurgy is the process of preparation of pure metal from its ore.

(c) The solids whose some physical properties like electrical resistance, refractive index, dielectric constant, etc. show different values in different directions in the same crystal, this property is called anisotropy.

Expression for maximum work:

When the volume of the gas increase by an infinitesimal amount dV in a single step, the small quantity of work

$$dW = -P_{\text{ex}} dV \text{ -----(i)}$$

As the expansion is reversible, the pressure of the gas is greater by a very small amount dP than P_{ex}

Hence,

$$\begin{aligned}
 P - P_{\text{ex}} &= dP \\
 \Rightarrow P_{\text{ex}} &= P - dP \text{ -----(ii)}
 \end{aligned}$$

Combining equation (i) and (ii), we get,

$$dW = -PdV \text{ -----(iii)}$$

$$\begin{aligned}
 \int_1^2 dW &= - \int_{V_1}^{V_2} PdV \\
 \Rightarrow W_{\text{max}} &= - \int_{V_1}^{V_2} PdV \text{ -----(iv)}
 \end{aligned}$$

Ideal gas equation for n moles of $PV = nRT$ or $P = \frac{nRT}{V}$

Substitution of equation (iv) into equation (iii) gives,

$$W_{\max} = - \int_{V_1}^{V_2} nRT \frac{dV}{V}$$

The elevation (ΔT_b) in the boiling point = 354.11-353.23 K

=0.88K

$$\begin{aligned} M_2 &= \frac{1000 \times W_2 \times K_b}{\Delta T_b \times W_1} \\ &= \frac{2.53 \text{Kkgmol}^{-1} \times 1.8 \text{g} \times 1000 \text{gkg}^{-1}}{0.88 \text{K} \times 90 \text{g}} \\ &= 58 \text{g mol}^{-1} \end{aligned}$$

Molar mass of the solute, $M_2 = 58 \text{g mol}^{-1}$

SECTION – II

Question 5:

Select and write the most appropriate answer from the given alternative for each sub-question:

(i) When primary amine reacts with CHCl_3 in alcoholic KOH, the product is :

- (a) aldehyde (b) alcohol (c) cyanide (d) an-isocyanide

(ii) $\text{CH}_3-\text{CH}_2-\text{Br} \xrightarrow[\Delta]{\text{Alco.KOH}} \text{B} \xrightarrow{\text{HBr}} \text{C} \xrightarrow{\text{Na/ether}} \text{D}$ the compound D is:

- (a) ethane (b) cancer (c) n-butane (d) n-pentane

(iii) Cisplatin compound is used in the treatment of :

- (a) malalaria (b) cancer (c) AIDS (d) yellow fever

(iv) A gas when passed through $\text{K}_2\text{Cr}_2\text{O}_7$ and dil. H_2SO_4 solution turns it green, the gas is:

- (a) CO_2 (b) NH_3 (c) SO_2 (d) Cl_2

(v) The alcohol used in thermometer is:

- (a) Methonal (b) Ethanol (c) Propanol (d) Butanol

(vi) Which of the following vitamins is the vitamin of alicyclic series?

- (a) Vitamin C (b) Vitamin K (c) Vitamin B (d) Vitamin A

(vii) Which of the following is the first oxidation product of secondary alcohol?

- (a) Alkene (b) Aldehyde (c) Ketone (d) Carboxylic acid

Solution 5:

(i) (d) an-isocyanide

(ii) (c) n- butane

(iii) (b) cancer

(iv) (c) SO₂

(v) Ethanol

(vi) (d) Vitamin A

(vii) (c) Ketone

Question 6: Answer any SIX of the following:

(i) How is diethyl ether prepared by continuous etherification process?

(ii) Write a note on Hofmann bromamide degradation.

(iii) How is ethanoic acid prepared from dry ice?

(iv) Write the molecular and structural formula of BHA and BHT.

(v) Explain the preparation of glucose from cane sugar.

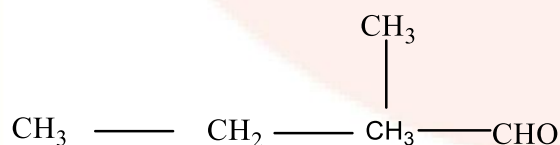
(vi) Write the factors which are related to the colour of transition metal ions.

(vii) Explain the following terms:

(a) Homopolymers (b) Elastomers

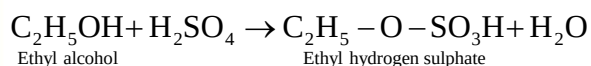
(viii) Define racemic mixture.

Give IUPAC name of

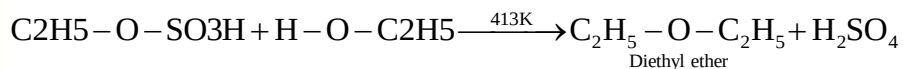


Solution 6:

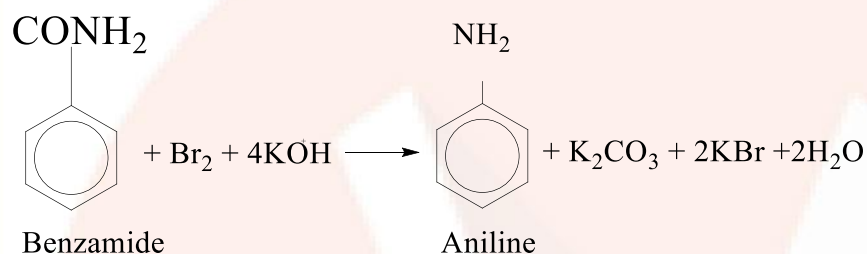
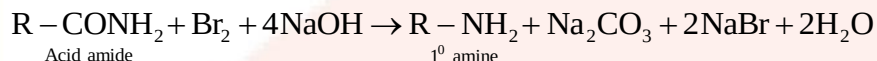
(i) Equal volumes of ethyl alcohol and conc. sulphuric acid are placed in a distillation flask. Ethyl alcohol reacts with conc. sulphuric acid to give ethyl hydrogen sulphate.



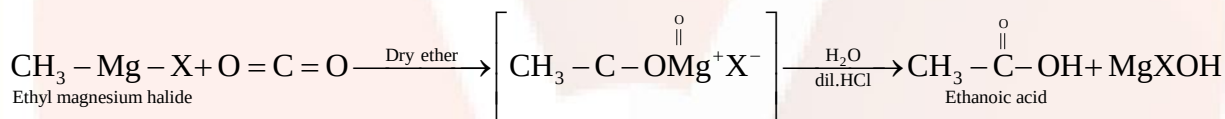
Then, excess ethyl alcohol is added to distillation flask. Ethyl hydrogen sulphate reacts with added ethyl alcohol to give diethyl ether.



(ii) **Hofmann bromamide reaction:** When a primary acid amide is heated with an aqueous or ethanolic, solution of NaOH or bromine, it gives primary amine with one carbon atom less, also known as Hoffmann bromamide reaction.



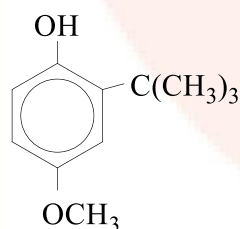
(iii) When Grignard reagent is added to solid carbon dioxide (dry ice), a magnesium salt of carboxylic acid is obtained which on acidification gives ethanoic acid.



(iv) Butylated Hydroxy Anisole(BHA):

Molecular formula: $\text{C}_{11}\text{H}_{16}\text{O}_2$

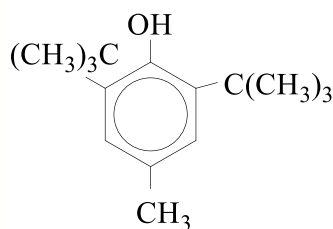
Structural formula:



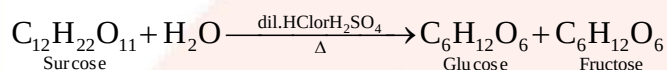
Butylated Hydroxy Toluene (BHT):

Molecular formula: $\text{C}_{15}\text{H}_{24}\text{O}$

Structural formula:



(v) When cane sugar or sucrose is boiled with dilute hydrochloric acid or sulphuric acid it undergoes hydrolysis to form glucose and fructose in equal amounts. While cooling, alcohol is added glucose being almost insoluble crystallize out first on cooling. Fructose being more soluble than glucose remains in the solution which can be separated by filtration.



(vi) Factors which are related to the color of transition metal ions:

(a) **Nature of the ligand:** Different ligands vary the energies of the d-orbitals of the metal ion. Few ligands, have stronger electrical fields resulting in a large energy gap when the d-orbitals splits into two group, while others produce much smaller gaps. Greater the splitting, greater amount of energy is needed to excite an electron from lower to higher orbitals.

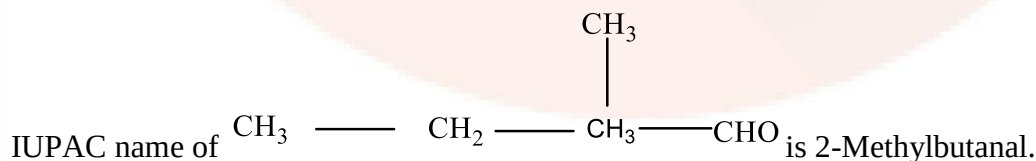
(b) **Oxidation state of Metals:** As the oxidation state of the metal increae, so also the amount of splitting of d- orbitals. Changes of oxidation state, changes the colour of the light absorbed and also the colour of light.

(c) **Coordination of ion:** The colour changes with the change of coordination.

(vi) (a) Homopolymers are the polymers whose repeating structural units are derived from only one type of monomer units. For example, plythene, Teflon.

(b) Elastomers are the polymers having the weakest intermolecular forces of attraction between the polymers chains. For example, Buna-S, Buna-N.

(viii) Racemic mixture is a mixture containing equal amounts of enantiomers which is optically inactive.



Question 7: Answer any THREE of the following:

(i) What is 'effective' atomic number' (EAN)?

Calculate the effective atomic number of the central metal atom in the following compounds:

(a) $K_4Fe(CN)_6$, $Fe(Z=26)$

(b) $Cr(CO)_6$, $Cr(Z=24)$

(ii) Write the different oxidation states of iron. Why +2 oxidation state of manganese is more stable? (Z of Mn = 25)

(iii) Write a note on 'aldol condensation'?

(iv) What are nucleic acids? Define complex lipids. Mention any 'two' functions of lipids.

Solution 7:

(i) Effective atomic number (EAN) is the total number of electrons around the central metal ion present in a complex.

(a) $K_4Fe(CN)_6$

$$EAN = Z - X + Y$$

$$= 26 - 2 + (6 \times 2)$$

$$= 26 - 2 + 12$$

$$= 36$$

(b) $Cr(CO)_6$

$$EAN = Z - X + Y$$

$$= 24 - 0 + 12$$

$$= 36$$

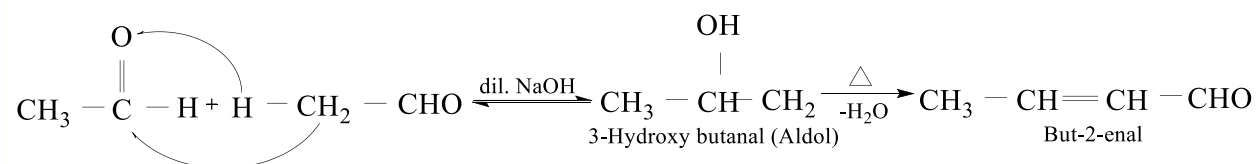
(ii) Iron exhibit stable oxidation state of +2 and +3 while +4, +5 and +6 are the unstable oxidation states of iron.

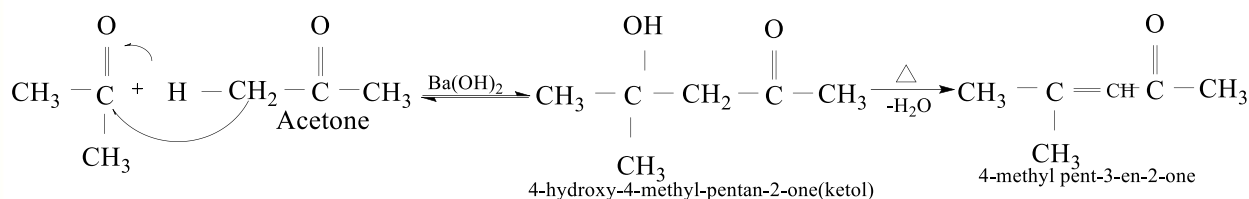
Electronic configuration of $Mn(Z=25) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^2$

Electronic configuration of $Mn^{2+}(Z=25) = 1s^2 2s^2 2p^6 3s^2 3p^6 3d^5 4s^0$

As half-filled orbital is more stable therefore +2 oxidation state of manganese is more stable.

(iii) **Aldol condensation:** Two molecule of aldehydes or ketones containing at least one α -hydrogen atom on treatment with dilute alkali undergo condensation to form β -hydroxy aldehydes (aldol) Or β -hydroxy ketones (ketol).





(iv) Nucleic acids are the biomolecules made of long chain polymers of nucleotides found in nucleus of cell which help in biosynthesis of protein and act as carries for transfer of hereditary characters.

Complex lipids are the esters of long chain fatty acids and can be hydrolyzed. They include triglycerides(animal fats and vegetable oils, Glycolipids, phospholipids and waxes..

Function of lipids:

- (a) Lipids are storage compounds, triglycerides serve as reserve energy of the body.
- (b) They regulate membrane permeability.
- (c) They act as electrical insulators to the nerve fibres and are also components of some enzyme systems.

Question 8: What is the action of mixture of NaNO_2 and dil. HCl on:

- (a) Ethyl amine, (b) Aniline, (c) Diethyl amine

How is nylon 6,6 prepared?

Write are 'antacids'?

Write any 'two' side effects of tranquilizers.

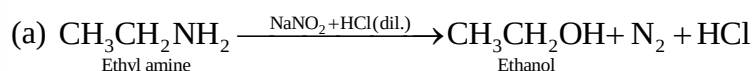
Or

Explain the mechanism of alkaline hydrolysis of tert-butyl bromide with energy profile diagram.

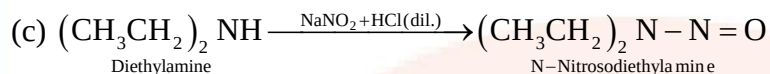
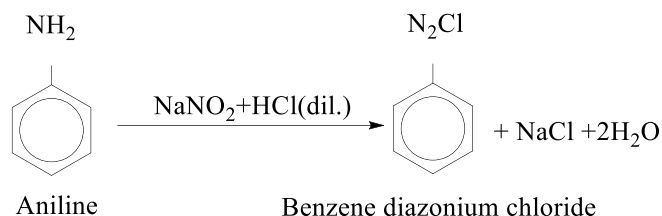
Define carbolic acid.

How carbolic acid is prepared from benzene sulphonic acid?

Solution 8:

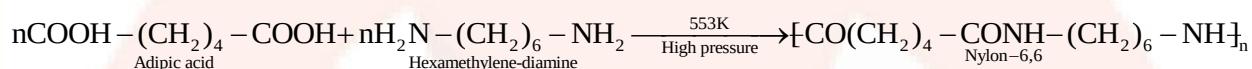


(b)



Preparation of Nylon 6,6:

It is prepared by the condensation polymerization of hexamethylene-diamine with adipic acid under high pressure and at high temperature.

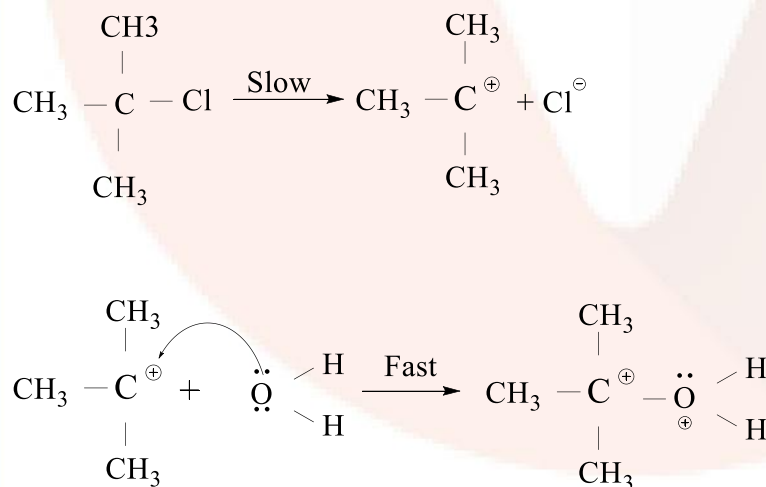


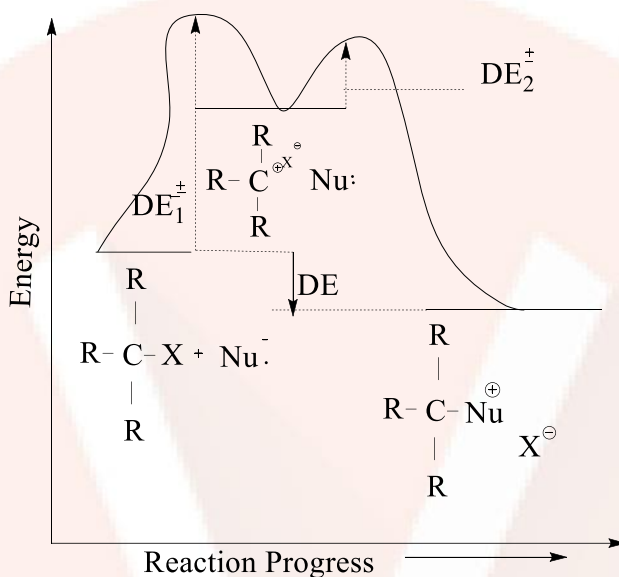
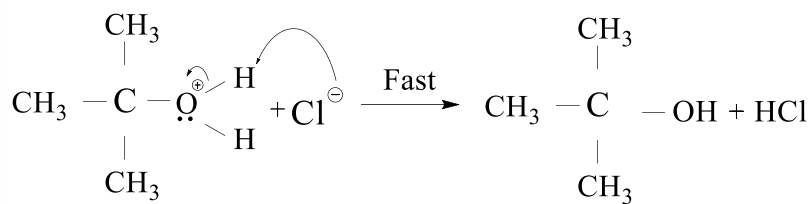
Antacids are the substances that remove the excess of acid and raise the pH of stomach to appropriate level. Example, magnesium hydroxide, sodium carbonate.

Side effects of tranquilizers are muscle spasms, irregular sleep patterns, restlessness, inability to relax and reparatory distress.

OR

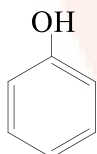
Mechanism of alkaline hydrolysis of tert-butyl bromide:





Carbolic acid is the simplest hydroxyl derivative of benzene. Phenol is also known as carbolic acid.

Structural formula:



Preparation of carbolic acid from benzene sulphonic acid:

Benzene sulphonic acid is fused with molten sodium hydroxide to give sodium phenoxide which on acidification gives phenol.

